CSE222/CSE505 Data Structures and Algorithms 2025 Spring Syllabus

Version: 2025.02.24.1 (There may be updates, check for the most recent version)

Meeting Times and Places

- **Lecture:** Wednesday, 15:30 – 17:30, Z10
- **PS/Lecture:** Monday 10:30 - 12:30, Z23
- **Lecture:** Thursday, 15:30 – 17:30, Z23

Instructor

- Gokhan Kaya
  - Email: gokhankaya@gtu.edu.tr
  - Room: 218
  - Office Hours: Wednesday 9:00-11:00

Teaching Assistants

- Enes Abdülhalik
- Arif Burak Dikmen

Management and Communication

- This course will be managed by an e-learning system: `MS Teams`. You will be added to the `Teams` page manually.
- You are responsible from all of the announcements made in class.
- In order to communicate, use your school email account only. The topic line should include `CSE222S25`.
- Do not send `Teams` chat messages to the instructor. They are going to be ignored.
- You may receive automated `Teams` chat messages related with your exam and/or assignment grades.

Textbooks

- Elliot Koffman, Paul Wolfgang, Data Structures: Abstraction and Design Using Java, 2E, Wiley, 2010.
- M. A. Weiss, Data Structures and Algorithm Analysis in C++, Addison Wesley, 2006.
- Cormen, Leiserton, Rivest, Introduction to Algorithms, MIT Press, 2001.

Prerequisites

- Solid C++ and Java programming skills are required
- A passing grade from CSE241 is required
  - If you do not satisfy the conditions , please talk to the instructor

Grading (Tentative)

- Items:
  - Midterm Exam (in the 8^{th} week of the semester):: `35%`
  - Final Exam: `35%`
  - Programming Assignments(~8 counts): `30%`
- `NA` and `VF` Conditions:
  - If the student does not submit **3** or more *Programming Assignment*, (s)he will not be admitted to midterm and final exams. The homework will be considered as "Not Submitted" if the student does not attend its demo session.
  - If the student's class attendance ratio is below the threshold stated in the attendance criteria(see *Attendance* section of this document), (s)he will get `NA`.
  - If the student does not enter the *midterm exam* (s)he will get `VF`.
  - If the student does not enter the *final exam* (s)he will get `VF` or `FF`.
  - If the student is dishonest about attendance, (s)he will get `NA` or `FF`.

Letter Grades

Total Points	Letter Grade
91-100	AA
85-90	BA
80-84	BB
70-79	CB
65-69	CC
55-64	DC
40-54	DD
<=39	FF

Attendance (Tentative)

- Attendance is required for the following students:
  - Students who failed this class with `NA`.
  - Students who take this class for the first time.
- Attendance will be taken in PS sessions.
- Attendance will be taken in class. (You have to attend at least 70% of the lectures)
- You are responsible from all the subjects covered in class.

Assignment Submission and Announcements

- All the class related announcements will be made **either** in class **or** on the class *teams* page.
- Students are required to read their emails regularly and check the *teams* page.
- The Programming Assignments will be announced at the *teams* page.
- The Programming Assignments will be submitted to the *teams* page.

Homework Submission and Announcements

- Some of the programming assignments will require student demonstrations. Every student is going to run his/her implementation and explain the solution.
- All the class related announcements will be made either in class or on the class team page.
- Students are required to read their emails regularly and check the team page.
- The Programming Assignments will be announced at the team page.
- The Programming Assignments will be submitted to the team page.

Cheating

- Copying someone's work (changing identifier names, order of the code etc...) is cheating.
- ChatGPT or any similar tool will be considered as a real person.
- Any trace of AI tools in the submissions may be considered as "cheating".
- If you cannot explain the submitted work, you submission may be considered as an act of "cheating".
- Cheating is not permitted.
- Do not cheat.
- **Do not sign the attendance sheet if you are not in class. (You will get `NA`, if detected)**

About The Course

- This course aims to introduce you some basic data structures and algorithms which are to be used as tools in designing solutions to problems.
- You will become familiar with the specification, usage, implementation and analysis of these data structures and algorithms.
- By the end of this course you should be able to:
  - Identify, implement and analyze the efficiency of the basic data structures introduced
  - Selected the most efficient data structure while designing algorithms to solve real-life problems

Topics to Be Covered

- Java Review
- Introduction to Algorithm Analysis
- Sequential Containers
- Stack
- Queue
- Recursion
- Trees
- Maps and Sets
- Graphs
- Self Balancing Trees
- Sorting